



Sadhana Devi S V 250701620-2025-CSE ▾

S2**Started on** Monday, 10 August 2026, 4:25 PM**State** Finished**Completed on** Monday, 10 August 2026, 5:57 PM**Time taken** 1 hour 32 mins**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeros(int a[], int left, int right)
4  {
5      if (left > right)
6          return 0;
7
8      if (a[left] == 0)
9          return right - left + 1;
10
11     int mid = (left + right) / 2;
12
13     return countZeros(a, mid + 1, right);
14 }
15
16 int main()
17 {
18     int m;
19     scanf("%d", &m);
20
21     int a[m];
22
23     for (int i = 0; i < m; i++)
24         scanf("%d", &a[i]);
25
26     printf("%d", countZeros(a, 0, m - 1));
27
28     return 0;
29 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

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S2**Started on** Monday, 10 August 2026, 5:57 PM**State** Finished**Completed on** Monday, 10 August 2026, 5:58 PM**Time taken** 55 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int majorityElement(int a[], int n)
4  {
5      int candidate = a[0];
6      int count = 1;
7
8      for (int i = 1; i < n; i++)
9      {
10         if (a[i] == candidate)
11             count++;
12         else
13             count--;
14
15         if (count == 0)
16         {
17             candidate = a[i];
18             count = 1;
19         }
20     }
21
22     return candidate;
23 }
24
25 int main()
26 {
27     int n;
28     scanf("%d", &n);
29
30     int a[n];
31
32     for (int i = 0; i < n; i++)
33         scanf("%d", &a[i]);
34

```

```
35     printf("%d", majorityElement(a, n));  
36  
37     return 0;  
38 }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 5:58 PM**State** Finished**Completed on** Monday, 10 August 2026, 5:59 PM**Time taken** 44 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, x;
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     scanf("%d", &x);
14
15     int low = 0, high = n - 1;
16     int floor = -1;
17
18     while (low <= high)
19     {
20         int mid = (low + high) / 2;
21
22         if (a[mid] <= x)
23         {
24             floor = a[mid];
25             low = mid + 1;
26         }
27         else
28         {
29             high = mid - 1;
30         }
31     }
32
33     printf("%d", floor);
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			

	Input	Expected	Got	
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Monday, 10 August 2026, 5:59 PM**State** Finished**Completed on** Monday, 10 August 2026, 6:00 PM**Time taken** 46 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int findPair(int a[], int left, int right, int x, int *e1, int *e2)
4  {
5      if (left >= right)
6          return 0;
7
8      if (a[left] + a[right] == x)
9      {
10         *e1 = a[left];
11         *e2 = a[right];
12         return 1;
13     }
14
15     if (a[left] + a[right] < x)
16         return findPair(a, left + 1, right, x, e1, e2);
17
18     return findPair(a, left, right - 1, x, e1, e2);
19 }
20
21 int main()
22 {
23     int n, x;
24     scanf("%d", &n);
25
26     int a[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &a[i]);
30
31     scanf("%d", &x);
32
33     int e1, e2;
34
35     if (findPair(a, 0, n - 1, x, &e1, &e2))
36     {
37         printf("%d\n", e1);
38         printf("%d\n", e2);
39     }
40     else
41     {
42         printf("No");
43     }
44
45     return 0;
46 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Sadhana Devi S V 250701620-2025-CSE** ▾**S2****Started on** Monday, 10 August 2026, 6:00 PM**State** Finished**Completed on** Monday, 10 August 2026, 6:08 PM**Time taken** 8 mins 6 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2
3  void quickSort(int a[], int low, int high)
4  {
5      if (low < high)
6      {
7          int i = low;
8          int j = high;
9          int pivot = a[low];
10
11         while (i < j)
12         {
13             while (a[i] <= pivot && i < high)
14                 i++;
15
16             while (a[j] > pivot)
17                 j--;
18
19             if (i < j)
20             {
21                 int temp = a[i];
22                 a[i] = a[j];
23                 a[j] = temp;
24             }
25         }
26
27         a[low] = a[j];
28         a[j] = pivot;
29
30         quickSort(a, low, j - 1);
31         quickSort(a, j + 1, high);
32     }
33 }
34
35 int main()
36 {
37     int n;
38     scanf("%d", &n);
39
40     int a[n];
41
42     for (int i = 0; i < n; i++)
43         scanf("%d", &a[i]);
44
45     quickSort(a, 0, n - 1);
46
47     for (int i = 0; i < n; i++)
48     {
49         printf("%d", a[i]);
50         if (i < n - 1)
51             printf(" ");
52     }

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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